

Mutation Analysis Using ClinVar

Description

This project involved exploring genetic variants in the **HBB (hemoglobin beta)** gene using ClinVar to investigate their potential clinical significance and understand how sequence changes may affect protein function.

Method

- Used ClinVar database for mutation analysis
- Searched for variants associated with the **HBB gene**
- Reviewed variant classification and molecular consequence
- Selected one variant for detailed interpretation

Results

Search results

Variation	Gene (Protein Change)	Type (Consequence)	Condition	Classification, Review status
☐ NM_000518.4(HBB):c.[20A>T;364G>A]	HBB, LOC107133510 +2 more (E122K +1 more)	Single nucleotide variant (missense variant)	Sickle cell-Hemoglobin O Arab disease	G Pathogenic ★
☐ NM_000518.4(HBB):c.364G>A (p.Glu122Lys)	HBB, LOC107133510 +1 more (E122K)	Single nucleotide variant (missense variant)	METHEMOGLOBINEMIA, BETA TYPE +11 more	G Pathogenic/Likely pathogenic ★★
☐ NM_000518.5(HBB):c.20A>T (p.Glu7Val)	HBB, LOC106099062 +1 more (E7V)	Single nucleotide variant (missense variant)	METHEMOGLOBINEMIA, BETA TYPE +15 more	G Pathogenic ★★
☐ NM_000518.4(HBB):c.19G>A (p.Glu7Lys)	HBB, LOC106099062 +1 more (E7K)	Single nucleotide variant (missense variant)	Hereditary persistence of fetal hemoglobin +16 more	G Pathogenic/Likely pathogenic ★★

Figure 1: Search results showing multiple HBB variants and their clinical classifications.

NM_000518.5(HBB):c.20A>T (p.Glu7Val)

CiteFollowPrintDownload

Germline

Classification

★ ★ ★ ★ ★

Pathogenic

53 out of 71 submissions contributed to this classification

Somatic

No data submitted for somatic clinical impact

Somatic

No data submitted for oncogenicity

Variant Details

This variant is part of 8 Haplotypes

- NM_000518.4(HBB):c.[20A>T;220G>A]
- NM_000518.4(HBB):c.[176C>G;20A>T]
- NM_000518.4(HBB):c.[20A>T;70G>A]
- NM_000518.4(HBB):c.[20A>T;364G>A]
- NM_000518.4(HBB):c.[20A>T;249G>Y]
- NM_000518.4(HBB):c.[20A>T;428C>T]
- NM_000518.4(HBB):c.[20A>T;271G>A]
- NM_000518.4(HBB):c.[205C>T;20A>T]

Identifiers:

NM_000518.5(HBB):c.20A>T (p.Glu7Val)

Type and length:

Variation ID: 15333 Accession: VCV000015333.177
single nucleotide variant, 1 bp

Figure 2: Detailed view of the selected HBB variant **NM_000518.5(HBB):c.20A>T (p.Glu7Val)** classified as pathogenic.

The selected mutation represents a **single nucleotide variant (SNV)** in the **HBB gene**, resulting in an amino acid substitution from **glutamic acid (Glu) to valine (Val)** at position 7 of the protein.

ClinVar classified this variant as **pathogenic**, indicating that published evidence supports its association with altered biological function and disease.

Interpretation

This analysis demonstrated how a small change at the DNA level can lead to a change in protein sequence and potentially influence biological outcomes.

Exploring ClinVar results also highlighted the importance of interpreting not only the mutation itself but also supporting evidence such as variant classification and molecular consequence.

Skills Demonstrated

- Mutation analysis
- Variant interpretation
- Clinical database navigation (ClinVar)
- Understanding genotype–protein relationships
- Bioinformatics data interpretation